

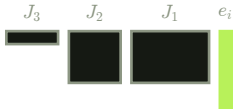
reverse mode: one sweep of vector-Jacobian products

$\bar{U}_3$ (1×4) J_2 (4×4) J_1 (4×6)



evaluate left to right: every partial product is a row vector
 $1 \times 1 \rightarrow 1 \times 4 \rightarrow 1 \times 4 \rightarrow 1 \times 6 =$ the whole gradient $\nabla_x L$

forward mode: one Jacobian-vector product per input



repeat for $i = 1 \dots 6$: six sweeps

right to left:

$6 \times 1 \rightarrow 4 \times 1$

$\rightarrow 4 \times 1 \rightarrow 1 \times 1$

$=$ one entry

$\partial L / \partial x_i$